# T-Test Simulator for Hypothesis Testing

**Application Documentation: Inputs, Outputs and Worked Test Cases**

## 1. Introduction

The T-Test Simulator is a web application that runs a t-test on data supplied by the user and explains the result. The user picks a test, supplies data, sets the direction of the alternative hypothesis and a significance level, and presses one button. The application returns the t-statistic, p-value, confidence interval, effect size, a decision about the null hypothesis, a written interpretation and several charts.

This document describes the application from the user's side: what goes in, what comes out and how to read it. It does not describe the code. All three tests were run in the application's own interface on the three sample datasets that come with it, and every screenshot is from those runs. Each test is shown with a two-tailed, a right-tailed and a left-tailed setting.

## 2. Using the Application

The sidebar on the left selects the test: *One-Sample t-Test*, *Independent 2-Sample* or *Paired Samples t-Test*. The workspace on the right holds, from top to bottom, the configuration panel, four result cards, the decision banner with the hypotheses and written interpretation, the charts, the assumption diagnostics and a log of the last ten runs. Two buttons in the top bar export the results and the data as CSV files.

### 2.1 Inputs

A switch at the top of the configuration panel chooses how data are supplied.

| Source | What the user does |
|---|---|
| Generate | Sets sample size, target mean, standard deviation, distribution shape and a random seed (and a pair correlation for paired tests). |
| Upload CSV | Chooses a .csv file, then picks which numeric column is Sample A and which is Sample B. |

| Source | What the user does |
|---|---|
| Manual Entry | Types or pastes numbers separated by commas, spaces or new lines. |

The hypothesis settings are the same for all three tests.

| Setting | Choices | Meaning |
|---|---|---|
| Alternative hypothesis ($H_1$) | Two-Sided, Left-Tailed, Right-Tailed | Direction of the test |
| Significance level (α) | 0.01, 0.05, 0.10 | Risk of a false rejection; also sets the confidence level (99%, 95%, 90%) |
| Hypothesized mean ($\mu_0$) | Any number | One-sample test only |
| Hypothesized difference ($\Delta_0$) | Any number, normally 0 | Independent and paired tests |
| Assume equal variances | Off (default) or on | Independent test only: off is Welch's test, on is Student's pooled test |

The form has no field for naming the groups, so results are labelled **Sample A** and **Sample B**. In this document Sample A is the first column of each file.

## 3. Outputs

| Output | How to read it |
|---|---|
| t-Statistic and df | How many standard errors the result lies from what $H_0$ predicts, and the degrees of freedom used |
| p-Value | Probability of a result at least this extreme if $H_0$ were true; shown as "< 0.0001" when tiny, and tagged significant when $p < \alpha$ |
| Mean Diff & CI | The observed difference and its confidence interval. In one-tailed tests one end of the interval is shown as $-\infty$ or $+\infty$ |

| Output | How to read it |
|---|---|
| Cohen's d and SE | Effect size (below 0.2 negligible, 0.2 to 0.5 small, 0.5 to 0.8 medium, 0.8 or more large) and the standard error |
| Decision banner | "Reject $H_0$" when $p < \alpha$, otherwise "Fail to reject $H_0$", with $H_0$ and $H_1$ written out |
| Interpretation and warnings | Plain-language reading, practical versus statistical significance, two cautions, and an amber box for small samples or non-normality |
| Charts | The t-curve with rejection regions, a histogram, group means with error bars and a descriptive table |
| Diagnostics | Shapiro-Wilk normality p-values and, for independent tests, Levene's equal-variance p-value |

For the one-sample test the interval beside "Mean Diff" is the interval for the population mean itself, while the figure above it is the sample mean minus $\mu_0$.

## 4. The Sample Data

Three CSV files come with the application and were loaded through **Upload CSV**.

| File and column | Test | n | Mean | Std. dev. |
|---|---|---|---|---|
| one_sample_exam_scores.csv: score | One-sample, $\mu_0 = 75$ | 20 | 84.28 | 5.57 |
| two_sample_drug_trial.csv: placebo | Independent, Sample A | 15 | 122.67 | 4.06 |
| two_sample_drug_trial.csv: treatment | Independent, Sample B | 15 | 111.67 | 4.25 |
| paired_weight_loss.csv: before_diet | Paired, Sample A | 12 | 86.48 | 6.10 |
| paired_weight_loss.csv: after_diet | Paired, Sample B | 12 | 83.62 | 5.50 |

## 5. One-Tailed and Two-Tailed Tests

The tail setting decides where the alternative hypothesis looks and where the rejection region sits on the t-curve.

| Setting | $H_1$ symbol | Question asked | Rejection region (α = 0.05) |
|---|---|---|---|
| Two-Sided | ≠ | Is there any difference? | Both tails, 0.025 each |
| Right-Tailed | > | Is the first quantity greater? | Right tail, 0.05 |
| Left-Tailed | < | Is the first quantity smaller? | Left tail, 0.05 |

A one-tailed test needs a smaller t to reject in its chosen direction, but it cannot detect an effect in the opposite direction: if t is large and positive and the test is left-tailed, the p-value is close to 1. The tail should therefore be chosen before looking at the data.

## 6. Case 1: One-Sample t-Test

**Scenario.** Is the average of these 20 exam scores different from 75? In the app, One-Sample t-Test is selected, Upload CSV is chosen, Sample A is the *score* column, $\mu_0$ is 75, α is 0.05 and the alternative is Two-Sided (Figure 1).

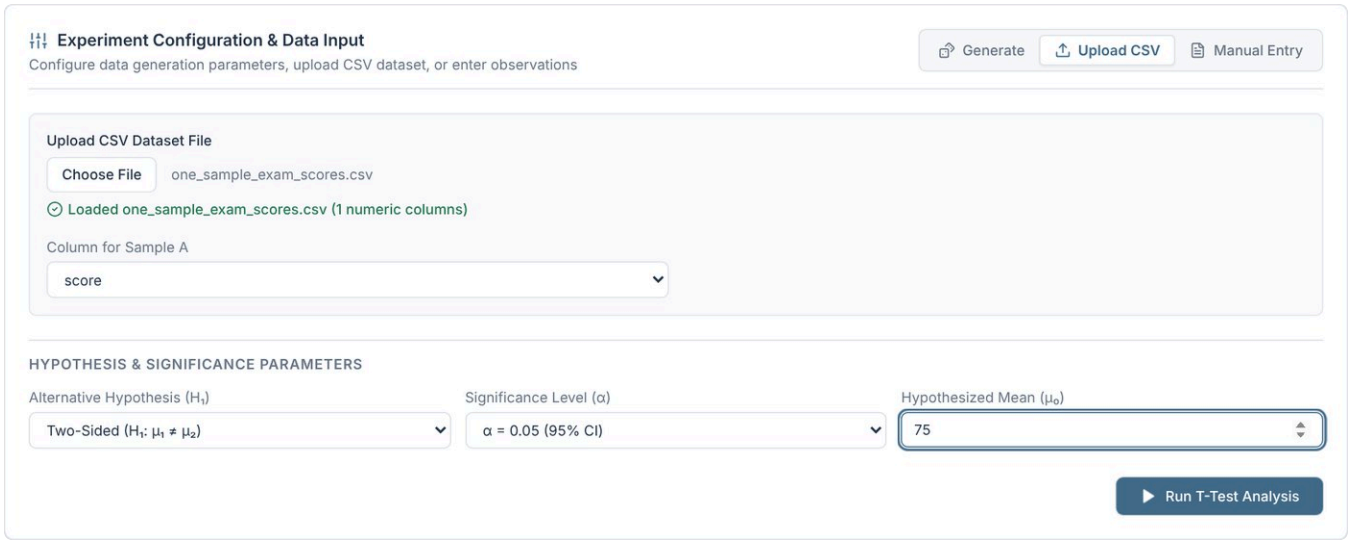



*Figure 1. One-sample test: configuration panel.*

**Two-tailed result (Figure 2).** The sample mean is 84.28, so the sample mean minus $\mu_0$ is 9.28. The t-statistic is 7.4515 with df = 19 and the p-value is below 0.0001. The 95% interval for the true mean, [81.67, 86.88], does not contain 75, and Cohen's d is 1.67 (Large). The banner reads Reject $H_0$ for $H_0$: μ(Sample A) = 75.00 against $H_1$: μ(Sample A) ≠ 75.00.




*Figure 2. One-sample test: two-tailed result cards, decision and hypotheses.*

**Charts (Figure 3).** The left chart is Student's t with df = 19. The red areas are the rejection regions beyond about ±2.093, and the legend gives the observed t of 7.45, far outside them. The right chart is a histogram of the 20 scores in 12 bars. The diagnostics panel shows a normality p-value of 0.9171, a pass, and no warning appears.




*Figure 3. One-sample test: t-distribution (left) and histogram (right).*

**One-tailed results (Figure 4).** The top result is the Right-Tailed run ($H_1$: μ > 75). The t-statistic is unchanged, the p-value is below 0.0001, the interval is (82.12, +∞) and $H_0$ is rejected. The bottom result is the Left-Tailed run ($H_1$: μ < 75). The scores lie above 75, the opposite direction, so p is 1.0000, the interval is (−∞, 86.43) and the decision is Fail to reject $H_0$.

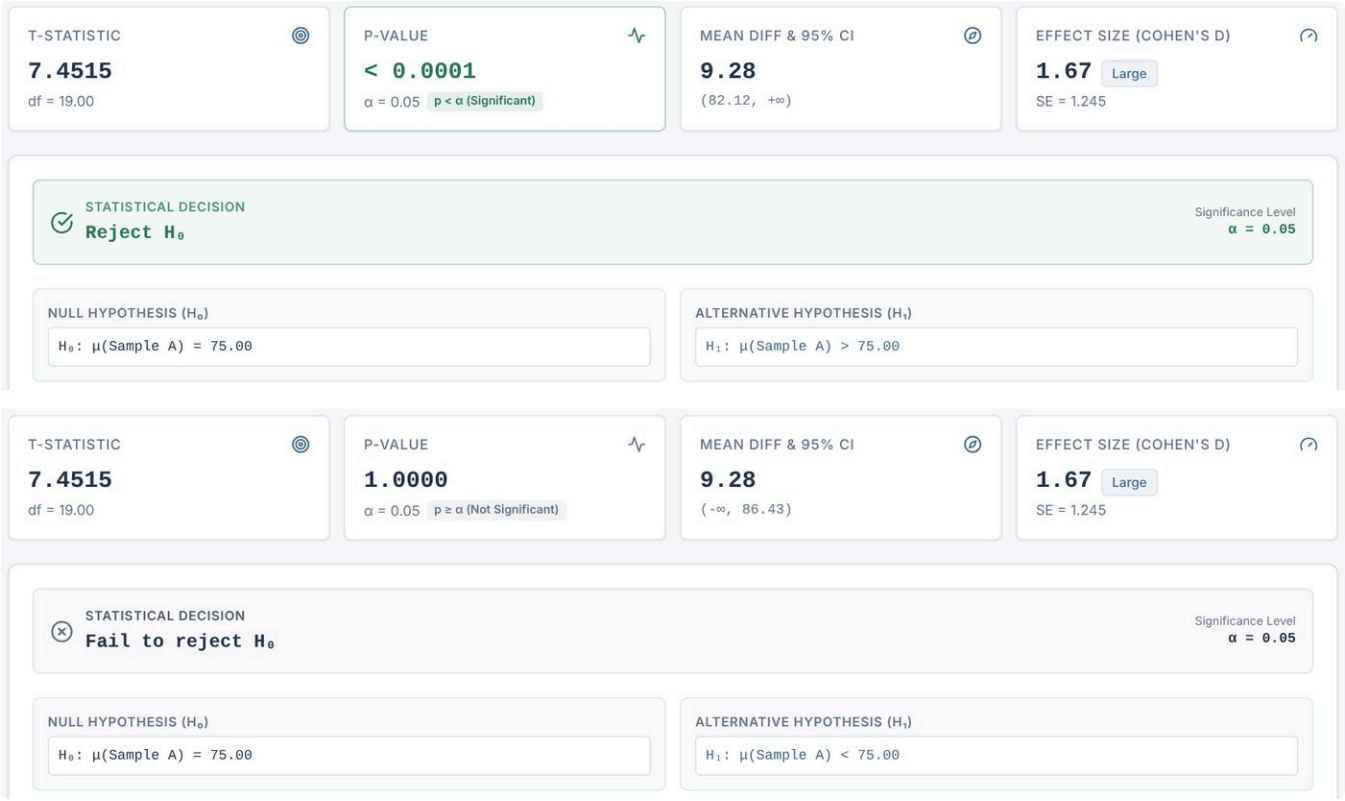



*Figure 4. One-sample test: right-tailed (top) and left-tailed (bottom) results.*

## 7. Case 2: Independent Two-Sample t-Test

**Scenario.** Fifteen patients took a placebo and a different fifteen took the treatment. Do the group means differ? In the app, Independent 2-Sample is selected, Sample A is *placebo*, Sample B is *treatment*, $\Delta_0$ is 0, α is 0.05 and *Assume Equal Variances* is left unchecked, so Welch's test is used (Figure 1).

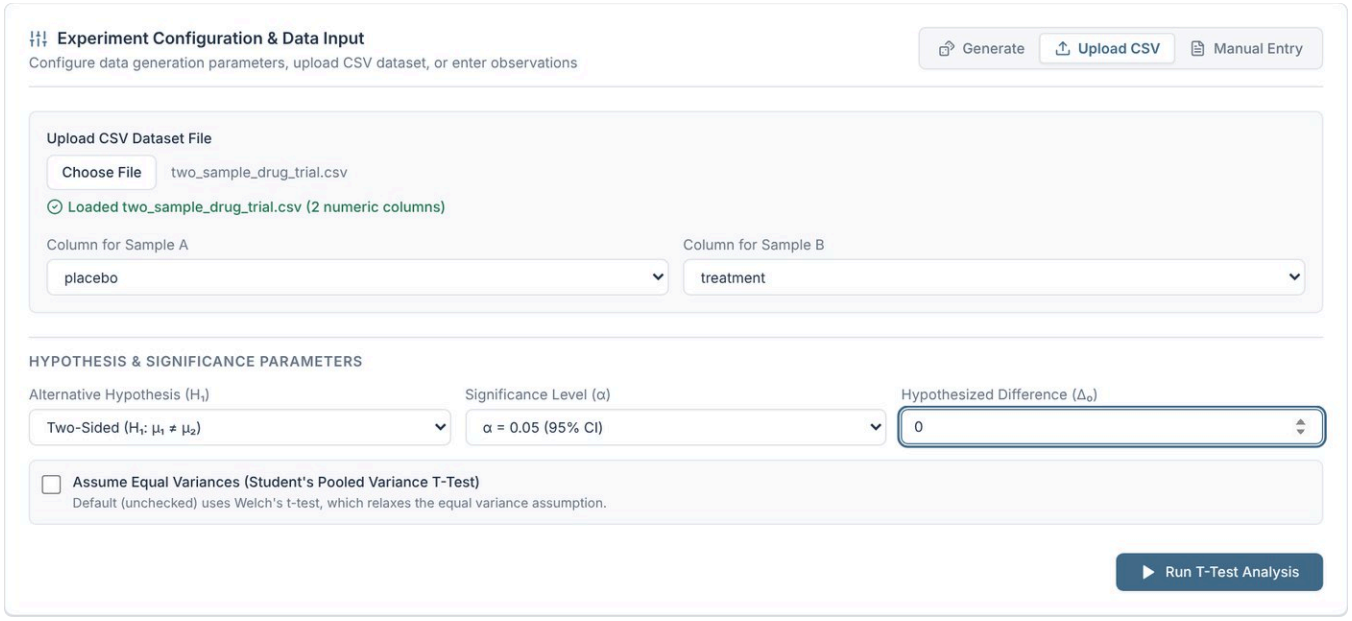



*Figure 1. Independent test: configuration panel.*

**Two-tailed result (Figure 2).** The means are 122.67 and 111.67, a difference of 11.00 with a standard error of 1.519. The t-statistic is 7.2407 with Welch's df = 27.94 and the p-value is below 0.0001. The 95% interval for the difference is [7.89, 14.11], which excludes zero, and Cohen's d is 2.64 (Large). The banner reads Reject $H_0$ for $H_0$: μ(Sample A) − μ(Sample B) = 0 against $H_1$: μ(Sample A) − μ(Sample B) ≠ 0.




*Figure 2. Independent test: two-tailed result cards, decision and hypotheses.*

**Charts (Figure 3).** The t-curve has df = 27.94 and its rejection regions begin at about ±2.049. The histogram draws the two groups in different colours, and they occupy different ranges, placebo near 115 to 130 and treatment near 105 to 119. The diagnostics panel shows Shapiro-Wilk p-values of 0.9996 and 0.9316 and a Levene p-value of 0.8237, all passes.

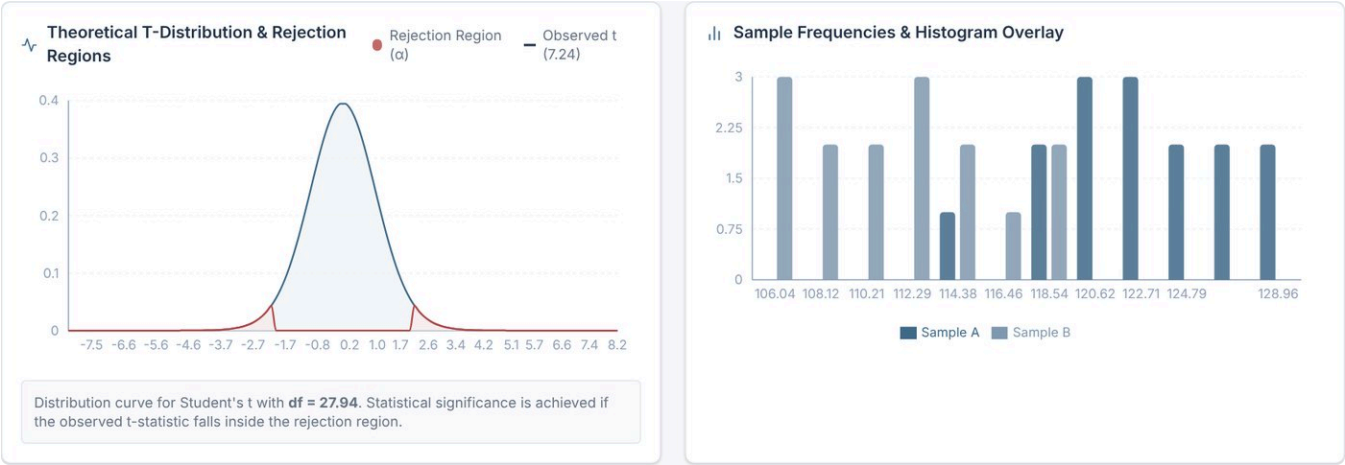



*Figure 3. Independent test: t-distribution (left) and histogram of both groups (right).*

**One-tailed results (Figure 4).** The Right-Tailed run ($H_1$: μA − μB > 0) rejects $H_0$ with an interval of (8.42, +∞). The Left-Tailed run ($H_1$: μA − μB < 0) gives p = 1.0000, an interval of (−∞, 13.58) and Fail to reject $H_0$, because placebo is the higher group. The difference is always Sample A minus Sample B, so swapping the columns would swap the two outcomes.

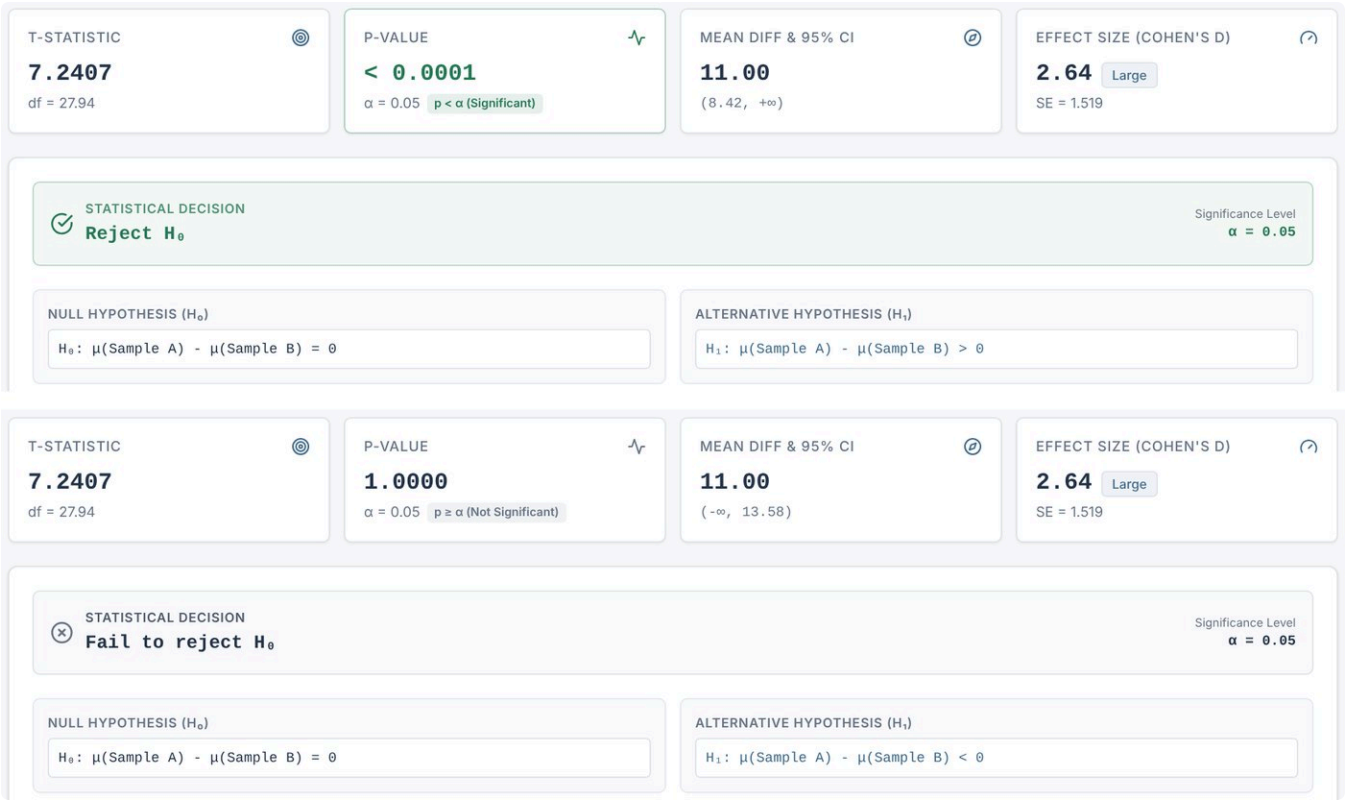



*Figure 4. Independent test: right-tailed (top) and left-tailed (bottom) results.*

**Welch or pooled.** Ticking *Assume Equal Variances* switches to Student's pooled test. On these data it gives the same t of 7.2407 with df = 28 and the same decision, because the group spreads and sizes are almost equal.

## 8. Case 3: Paired t-Test

**Scenario.** Twelve people were weighed before and after a diet, so each row is one person and the two columns are dependent. In the app, Paired Samples t-Test is selected, Sample A is *before_diet*, Sample B is *after_diet*, $\Delta_0$ is 0, $\alpha$ is 0.05 and the alternative is Two-Sided (Figure 1).

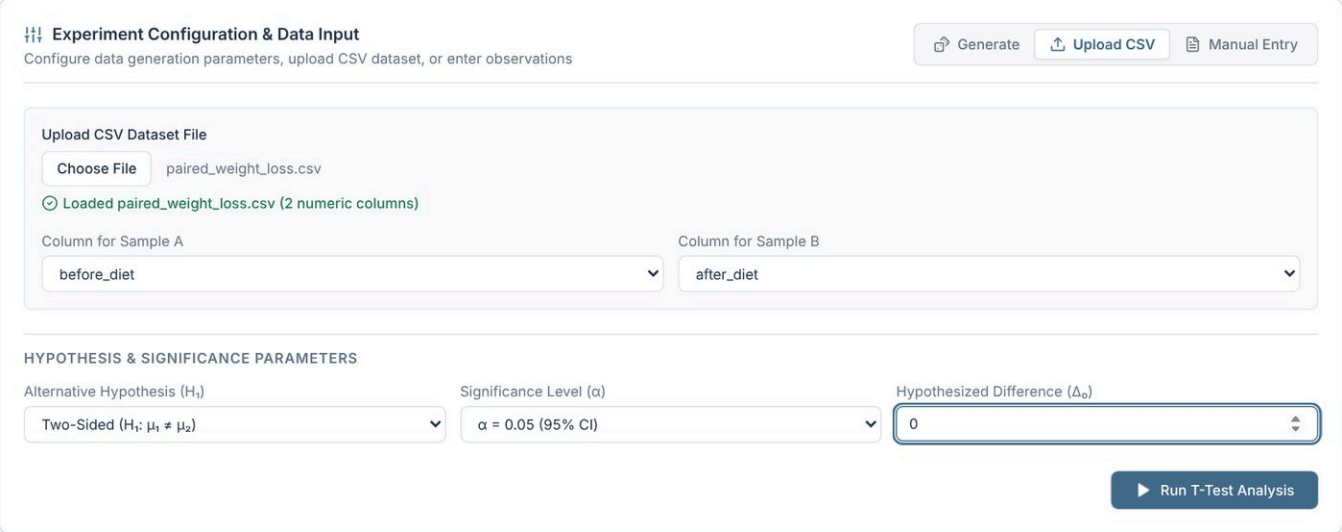



*Figure 1. Paired test: configuration panel.*

**Two-tailed result (Figure 2).** The app subtracts each pair (Sample A minus Sample B) and tests the twelve differences as one sample. The mean difference is 2.87 kg with a standard error of 0.191. The t-statistic is 15.0225 with df = 11 and the p-value is below 0.0001. The interval [2.45, 3.29] is entirely above zero and Cohen's d is 4.34 (Large). The banner reads Reject $H_0$ for $H_0$: μ_d(Sample A − Sample B) = 0 against $H_1$: μ_d(Sample A − Sample B) ≠ 0. Because the differences are all in the same direction, t is very large for a loss of under 3 kg.




*Figure 2. Paired test: two-tailed result cards, decision and hypotheses.*

**Charts (Figure 3).** The t-curve has df = 11 and its rejection regions begin at about ±2.201. The histogram overlays the before and after weights. An amber warning box under the hypotheses says that the Sample A size is small (n = 12, below 15) and that the normality assumption is critical. This is a caution, not an error, and the run still completes.

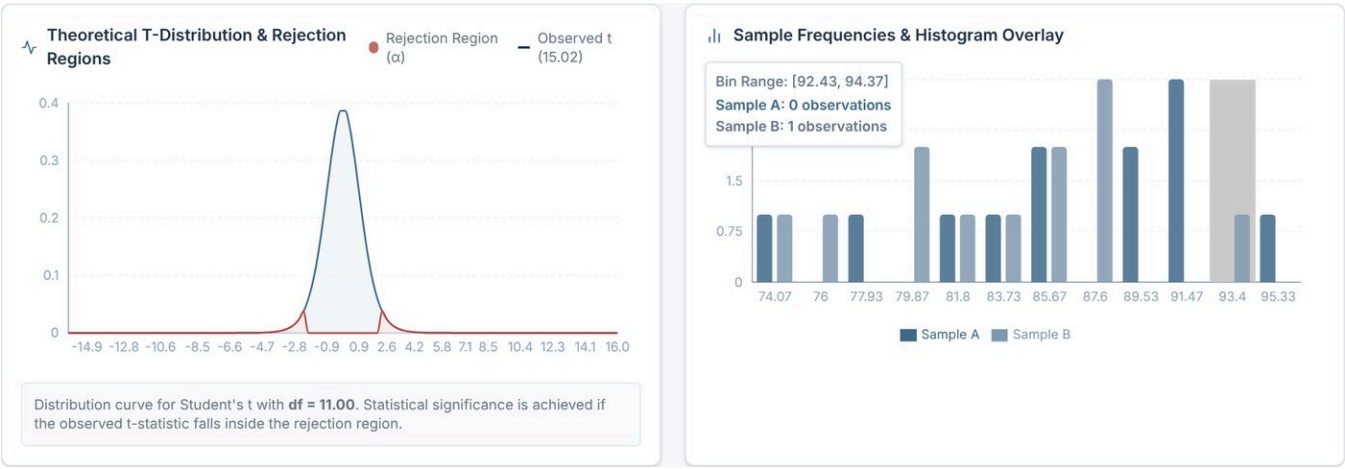



*Figure 3. Paired test: t-distribution (left) and histogram of both measurements (right).*

**One-tailed results (Figure 4).** The Right-Tailed run ($H_1$: μ_d > 0) asks whether weight before exceeds weight after, the natural question for a diet study. It rejects $H_0$ with an interval of (2.52, +∞). The Left-Tailed run ($H_1$: μ_d < 0) gives p = 1.0000, an interval of (−∞, 3.21) and Fail to reject $H_0$, because nobody gained weight. A positive mean difference means weight went down.

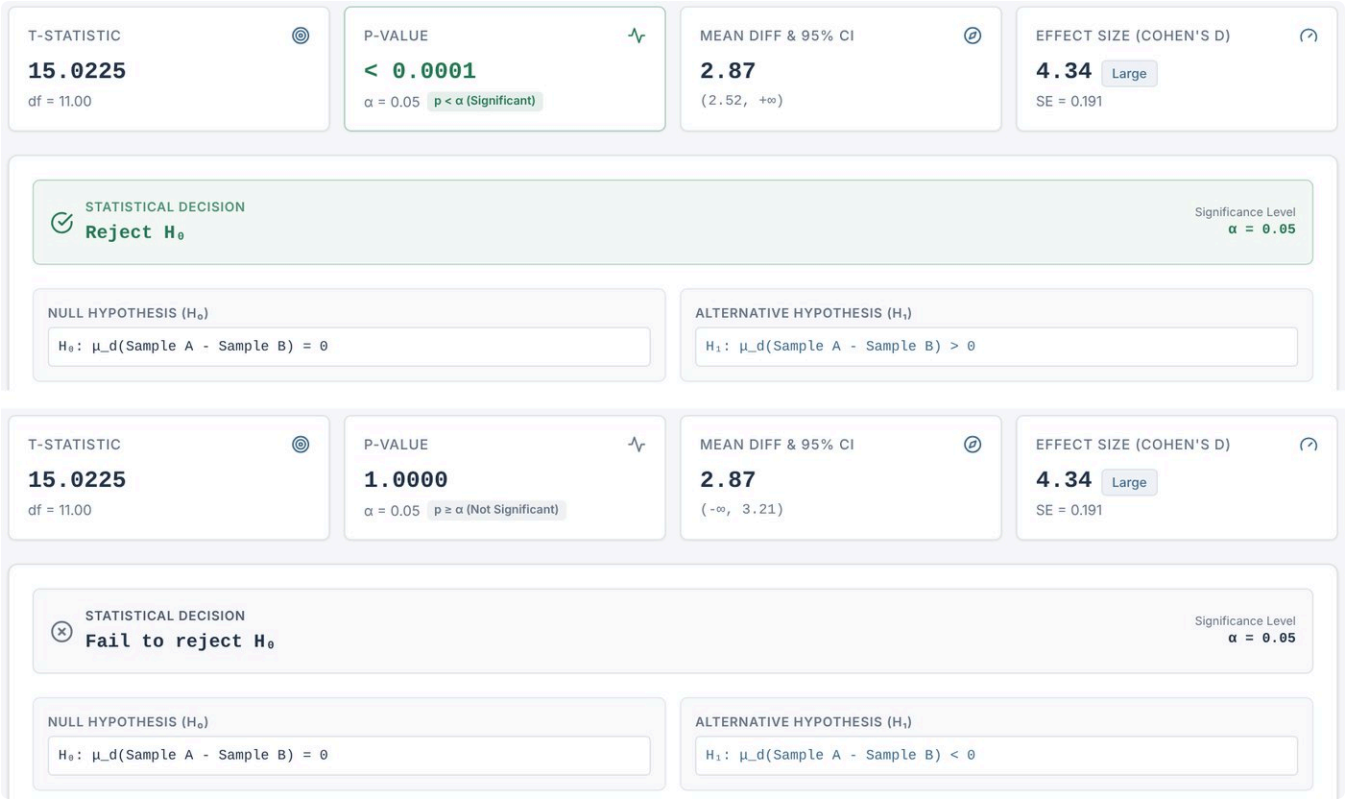



*Figure 4. Paired test: right-tailed (top) and left-tailed (bottom) results.*

## 9. Summary of the Nine Runs

| Test | Tail | t | df | p shown | Decision |
|---|---|---|---|---|---|
| One-sample | Two-tailed | 7.4515 | 19 | < 0.0001 | Reject $H_0$ |
| One-sample | Right-tailed | 7.4515 | 19 | < 0.0001 | Reject $H_0$ |
| One-sample | Left-tailed | 7.4515 | 19 | 1.0000 | Fail to reject $H_0$ |
| Independent | Two-tailed | 7.2407 | 27.94 | < 0.0001 | Reject $H_0$ |
| Independent | Right-tailed | 7.2407 | 27.94 | < 0.0001 | Reject $H_0$ |
| Independent | Left-tailed | 7.2407 | 27.94 | 1.0000 | Fail to reject $H_0$ |
| Paired | Two-tailed | 15.0225 | 11 | < 0.0001 | Reject $H_0$ |
| Paired | Right-tailed | 15.0225 | 11 | < 0.0001 | Reject $H_0$ |
| Paired | Left-tailed | 15.0225 | 11 | 1.0000 | Fail to reject $H_0$ |

In every family the t-statistic does not change with the tail setting. Only the p-value, the interval, the rejection region and the decision change. The two-tailed and right-tailed runs reject $H_0$ because the data point to the right, and the left-tailed run fails to reject because

they point the other way.

## 10. Conclusion

On the three sample datasets the application gives consistent, readable answers for all nine combinations of test and tail. The exam scores are significantly above 75, the placebo readings are significantly higher than the treatment readings, and the diet produced a significant average loss of about 2.87 kg. A user should choose the test from the way the data were collected, choose the tail from the research question before seeing the numbers, read the p-value together with the interval and Cohen's d, and check the diagnostics before trusting the conclusion.